

CHAPTER

3.4

AMPLIFIERS

1. If the transistor parameter are $\beta = 180$ and Early voltage $V_A = 140$ V and it is biased at $I_{CQ} = 2$ mA, the values of hybrid- π parameter g_m , r_π and r_o are respectively

- (A) 14 A/V, 2.33 k Ω , 90 k Ω
- (B) 14 A/V, 90 k Ω , 2.33 k Ω
- (C) 77 mA/V, 2.33 k Ω , 70 k Ω
- (D) 77.2 A/V, 70 k Ω , 2.33 k Ω

Statement for Q.2–3.

Consider the circuit of fig. P3.4.2–3. The transistor parameters are $\beta = 120$ and $V_A = \infty$.

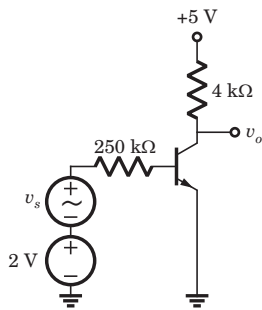


Fig. P3.4.2–3

2. The hybrid- π parameter values of g_m , r_π and r_o are

- (A) 24 mA/V, ∞ , 5 k Ω
- (B) 24 mA/V, 5 k Ω , ∞
- (C) 48 mA/V, 10 k Ω , 18.4 k Ω
- (D) 48 mA/V, 18.4 k Ω , 10 k Ω

3. The small signal voltage gain $A_v = v_o/v_s$ is

- (A) -4.38
- (B) 4.38
- (C) -1.88
- (D) 1.88

4. The nominal quiescent collector current of a transistor is 1.2 mA. If the range of β for this transistor is $80 \leq \beta \leq 120$ and if the quiescent collector current changes by ± 10 percent, the range in value for r_π is

- (A) 1.73 k Ω < r_π < 2.59 k Ω
- (B) 1.93 k Ω < r_π < 2.59 k Ω
- (C) 1.73 k Ω < r_π < 2.59 k Ω
- (D) 1.56 k Ω < r_π < 2.88 k Ω

Statement for Q.5–6:

Consider the circuit in fig. P3.4.5.6. The transistor parameter are $\beta = 100$ and $V_A = \infty$.

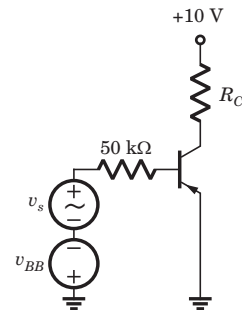


Fig. P3.4.5–6

5. If Q-point is in the center of the load line and $I_{CQ} = 0.5$ mA, the values of V_{BB} and R_C are

- (A) 10 k Ω , 0.95 V
- (B) 10 k Ω , 1.45 V
- (C) 48 k Ω , 0.95 V
- (D) 48 k Ω , 1.45 V

Statement for Q.14–15:

Consider the common Base amplifier shown in fig. P3.4.14–15. The parameters are $g_m = 2 \text{ mS}$ and $r_o = 250 \text{ k}\Omega$. Find the Thevenin equivalent faced by load resistance R_L .

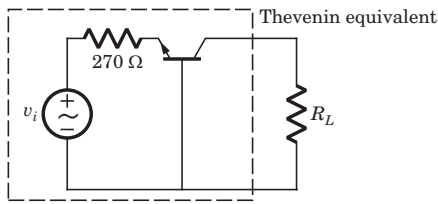


Fig. P3.4.14–15

14. The Thevenin voltage v_{TH} is

- (A) $263v_i$ (B) $132v_i$
(C) $346v_i$ (D) $498v_i$

15. The Thevenin equivalent resistance R_{TH} is

- (A) $384 \text{ k}\Omega$ (B) $697 \text{ k}\Omega$
(C) $408 \text{ k}\Omega$ (D) $915 \text{ k}\Omega$

Statement for Q.16–17:

The common-base amplifier is drawn as a two-port in fig. P3.4.16–17. The parameters are $\beta = 100$, $g_m = 3 \text{ mS}$, and $r_o = 800 \text{ k}\Omega$.

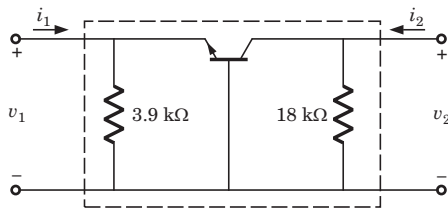


Fig. P3.4.16–17

16. The h-parameter h_{21} is

- (A) 2.46 (B) 0.9
(C) 0.5 (D) 0.67

17. The h-parameter h_{12} is

- (A) 3.8×10^{-4} (B) 4.83×10^{-3}
(C) 3.8×10^4 (D) 4.83×10^3

18. For an n -channel MOSFET biased in the saturation region, the parameters are $K_n = 0.5 \text{ mA/V}^2$, $V_{TN} = 0.8 \text{ V}$ and $\lambda = 0.01 \text{ V}^{-1}$, and $I_{DQ} = 0.75 \text{ mA}$. The value of g_m and r_o are

- (A) 0.68 mS , $603 \text{ k}\Omega$ (B) 1.22 mS , $603 \text{ k}\Omega$
(C) 1.22 mS , $133 \text{ k}\Omega$ (D) 0.68 mS , $133 \text{ k}\Omega$

19. For an n -channel MOSFET biased in the saturation region, the parameters are $V_{TN} = 1 \text{ V}$, $\frac{1}{2}\mu_n C_{ox} = 18 \text{ mA/V}^2$ and $\lambda = 0.015 \text{ V}^{-1}$ and $I_{DQ} = 2 \text{ mA}$. If transconductance is $g_m = 3.4 \text{ mA/V}$, the width-to-length ratio is

- (A) 80.6 (B) 43.2
(C) 190 (D) 110

20. In the circuit of fig. P3.4.20, the parameters are $g_m = 1 \text{ mA/V}$, $r_o = 50 \text{ k}\Omega$. The gain $A_v = v_o/v_s$ is

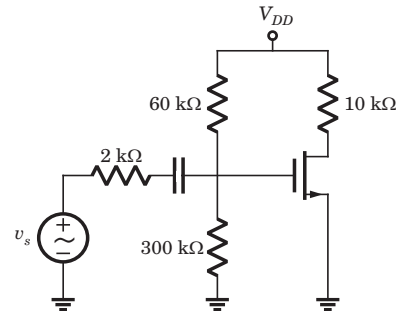


Fig. P3.4.20

- (A) -8.01 (B) 8.01
(C) 14.16 (D) -14.16

Statement for Q.21–23:

For the circuit shown in fig. P3.4.21–23 transistor parameters are $V_{TN} = 2 \text{ V}$, $K_n = 0.5 \text{ mA/V}^2$ and $\lambda = 0$. The transistor is in saturation.

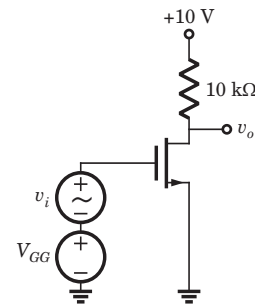


Fig. P3.4.21–23

21. If I_{DQ} is to be 0.4 mA , the value of V_{GSQ} is

- (A) 5.14 V (B) 4.36 V
(C) 2.89 V (D) 1.83 V

22. The values of g_m and r_o are

- (A) 0.89 mS , ∞ (B) 0.89 mS , 0
(C) 1.48 mS , 0 (D) 1.48 mS , ∞

23. The small signal voltage gain A_v is

- (A) 14.3 (B) -14.3
(C) -8.9 (D) 8.9

- (A) 4.44
(C) 2.22

- (B) -4.44
(D) -2.22

Statement for Q.33–34:

Consider the source-follower circuit in fig. P3.4.33-34. The values of parameter are $g_m = 2 \text{ mS}$ and $r_o = 100 \text{ k}\Omega$.

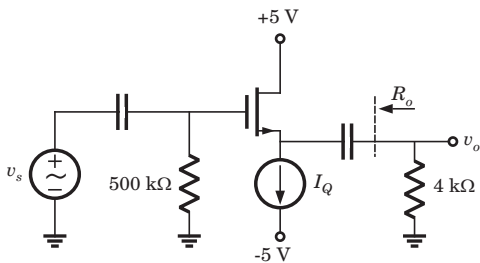


Fig. P3.4.33-34

33. The voltage gain A_v is

- (A) 0.89
(C) 2.79
- (B) -0.89
(D) -2.79

34. The output resistance R_o is

- (A) 100 kΩ
(C) 1.33 kΩ
- (B) 0.498 kΩ
(D) None of the above

SOLUTIONS

$$1. (C) \quad g_m = \frac{I_{CQ}}{V_t} = \frac{2\text{m}}{0.0259} = 77.2 \text{ mA/V}$$

$$r_\pi = \frac{\beta V_t}{I_{CQ}} = \frac{\beta}{g_m} = \frac{180}{77.2\text{m}} = 2.33 \text{ k}\Omega$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{140}{2\text{m}} = 70 \text{ k}\Omega$$

$$2. (B) \quad I_{BQ} = \frac{2 - 0.7}{250\text{k}} = 5.2 \text{ }\mu\text{A}$$

$$I_{CQ} = \beta I_B = (120)(5.2\mu) = 0.624 \text{ mA}$$

$$g_m = \frac{I_{CQ}}{V_t} = \frac{0.624}{0.0259} = 24 \text{ mA/V}$$

$$r_\pi = \frac{\beta V_t}{I_{CQ}} = \frac{\beta}{g_m} = \frac{120}{24\text{m}} = 5 \text{ k}\Omega, \quad r_o = \infty$$

$$3. (C) \quad A_v = -g_m R_C \left(\frac{r_\pi}{r_\pi + R_B} \right) = \frac{\beta R_C}{r_\pi + R_B}$$

$$= -(24\text{m})(4\text{k}) \left(\frac{5\text{k}}{5\text{k} + 250\text{k}} \right) = -1.88$$

$$4. (D) \quad r_\pi = \frac{\beta V_T}{I_{CQ}}$$

$$r_{\pi(max)} = \frac{(120)(0.0259)}{1.08\text{m}} = 2.88 \text{ k}\Omega,$$

$$r_{\pi(min)} = \frac{(80)(0.0259)}{1.32\text{m}} = 1.56 \text{ k}\Omega$$

$$5. (A) \quad V_{ECQ} = \frac{1}{2} V_{CC} = 5 \text{ V}$$

$$V_{ECQ} = 10 - I_{CQ} R_C = 5$$

$$\Rightarrow 10 - (0.5\text{m}) R_C = 5$$

$$R_C = 10 \text{ k}\Omega,$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.5}{100} = 5 \text{ }\mu\text{A}$$

$$V_{EB(ON)} + I_{BQ} R_B = V_{BB}$$

$$\Rightarrow 0.7 + (5\mu)(50\text{k}) = 0.95 \text{ V}$$

$$6. (D) \quad g_m = \frac{I_{CQ}}{V_t} = \frac{0.5}{0.0259} = 19.3 \text{ mA/V}$$

$$r_\pi = \frac{\beta V_t}{I_{CQ}} = \frac{(100)(0.0259)}{0.5\text{m}} = 5.18 \text{ k}\Omega, \quad r_o = \infty$$

$$7. (B) \quad I_{CQ} = \left(\frac{100}{1001} \right) (0.35) = 0.347 \text{ mA}$$

The small-signal equivalent circuit is as shown in fig. S3.4.7

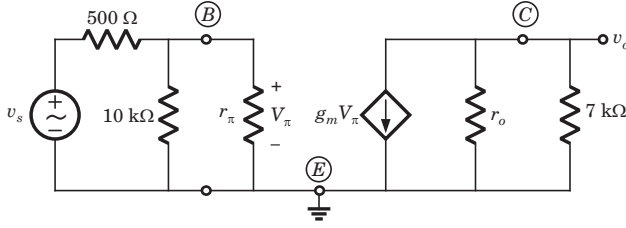


Fig. S3.4.7

$$V_{\pi} = \frac{r_{\pi} \parallel 10k}{500 + r_{\pi} \parallel 10k} (v_s), v_o = -g_m V_{\pi} (r_o \parallel 7k)$$

$$\frac{v_o}{v_s} = -g_m \left(\frac{r_{\pi} \parallel 10k}{500 + r_{\pi} \parallel 10k} \right) (r_o \parallel 7k)$$

$$g_m = \frac{I_{CQ}}{V_t} = \frac{0.347m}{0.0259} = 13.13 \text{ mA/V}$$

$$r_{\pi} = \frac{\beta}{g_m} = \frac{\beta V_t}{I_{CQ}} = \frac{100}{13.13m} = 7.6 \text{ k}\Omega$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{100}{0.347m} = 288 \text{ k}\Omega$$

$$r_o \parallel 7k = \frac{288 \times 7}{288 + 7} = 6.83 \text{ k}\Omega, r_{\pi} \parallel 10k = \frac{7.6 \times 10}{7.6 + 10} = 4.32 \text{ k}\Omega$$

$$A_v = -13.13m \left(\frac{4.32k}{500 + 4.32k} \right) (6.83k) = -80$$

8. (C) DC Analysis: $I_{CQ} = I_{EQ}$

$$V_{CEQ} = 5 = 10 - I_{CQ} (R_C + R_E)$$

$$\Rightarrow 5 = 10 - I_{CQ} (1.2k + 0.2k) \Rightarrow I_{CQ} = 3.57 \text{ mA}$$

$$I_{BQ} = \frac{3.57}{150} = 23.8 \text{ }\mu\text{A}$$

AC Analysis:

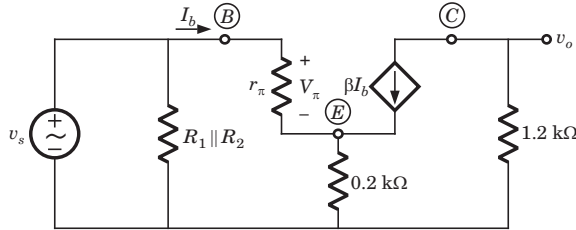


Fig. S3.4.8

$$r_{\pi} = \frac{\beta V_t}{I_{CQ}} = (150) \frac{(0.0259)}{3.57m} = 1.09 \text{ k}\Omega, r_o = \infty$$

$$A_v = \frac{v_o}{v_s} = \frac{-(\beta I_b) R_C}{v_s}, v_s = I_b r_{\pi} + (\beta + 1) R_E I_b$$

$$A_v = \frac{-\beta R_C}{r_{\pi} + (1 + \beta) R_E} = \frac{-(150)(12k)}{1.09k + (151)(0.2k)} = -5.75$$

9. (A) DC Analysis: $V_{TH} = \frac{50}{10 + 50} (12) = 10 \text{ V}$

$$R_{TH} = 11 \parallel 50 = 8.33 \text{ k}\Omega$$

$$I_{BQ} = \frac{12 - 0.7 - 10}{8.33k + (101)1k} = 11.9 \text{ }\mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = 1.19 \text{ mA}, I_{EQ} = 12 \text{ mA}$$

$$V_{ECQ} = 12 - (1.20)1 - (1.19)2 = 8.42 \text{ V}$$

AC Analysis:

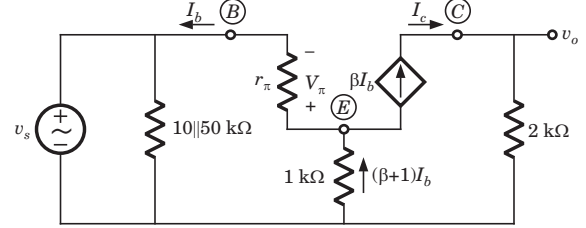


Fig. S3.4.9

$$r_{\pi} = \frac{\beta V_t}{I_{CQ}} = \frac{(100)(0.0259)}{1.19m} = 2.18 \text{ k}\Omega$$

$$v_o = \beta I_b (2k), v_s = -(\beta + 1) I_b (1k) + I_b (r_{\pi})$$

$$A_v = \frac{v_o}{v_s} = \frac{-\beta(2k)}{r_{\pi} + (\beta + 1)1k} = \frac{-(100)(2k)}{2.18k + (100)(1k)} = -1.96$$

10. (B) $V_{ECQ} = 8.42 \text{ V}$,

$$\text{For } 1 \leq v_{EC} \leq 11 \text{ V}, \Delta v_{EC} = 11 - 8.42 = 2.58 \text{ V}$$

$\Rightarrow$ Output voltage swing = 5.16 V peak to peak.

11. (B) Since the B-C junction is not reverse biased, the transistor continues to operate in the forward-active-mode

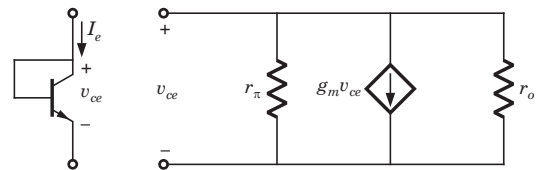


Fig. S 3.4.11

$$r = \frac{v_{ce}}{g_m V_{ce}} = \frac{1}{g_m}, \text{ So } r_{\pi} \parallel \left(\frac{1}{g_m} \right) \parallel r_o$$

$$r_{\pi} = \frac{(100)(0.0259)}{2m} = 2.33 \text{ k}\Omega$$

$$g_m = \frac{I_{CQ}}{V_t} = \frac{2m}{0.0259} = 77.2 \text{ mA/V}$$

$$\frac{1}{g_m} = 12.95, r_o = \frac{V_A}{I_{CQ}} = \frac{150}{2m} = 75 \text{ k}\Omega$$

$$r_e = (2.33k) \parallel (12.95) \parallel (75k) = 12.87 \text{ }\Omega$$

$$12. (C) r_o = \frac{V_A}{I_{CQ}} \Rightarrow I_{CQ} = \frac{V_A}{r_o} = \frac{75}{200k} = 0.375 \text{ mA}$$

$$13. (B) r_\pi = \frac{\beta V_t}{I_{CQ}} = \frac{75(0.0259)}{1m} = 1.94 \text{ k}\Omega$$

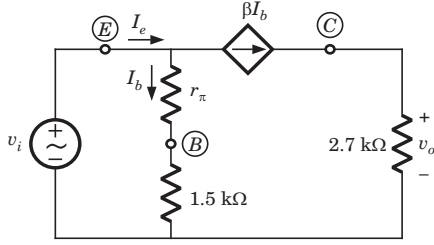


Fig. S 3.4.13

$$v_i = I_b(r_\pi + 15k), I_{in} = I_e = (\beta + 1)I_b$$

$$R_{in} = \frac{V_i}{I_e} = \frac{(r_\pi + 15k)}{(\beta + 1)} = \frac{1.94 + 15k}{76} = 45 \Omega$$

14. (D) The equivalent circuit is shown below

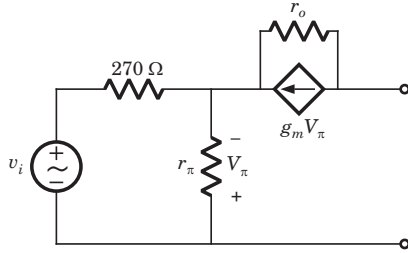


Fig. S 3.4.14

$$\text{Removing the } R_L, -V_\pi = \frac{v_i r_\pi}{270 + r_\pi}$$

$$v_{TH} = -r_o g_m V_\pi - V_\pi = \frac{v_i r_\pi (1 + g_m r_o)}{270 + r_\pi}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{2m} = 50 \text{ k}\Omega$$

$$v_{TH} = \frac{v_i 50k(1 + (2m)(250k))}{270 + 50k} = 498 v_i$$

15. (A) The equivalent small-signal circuit is shown in fig. S3.4.15

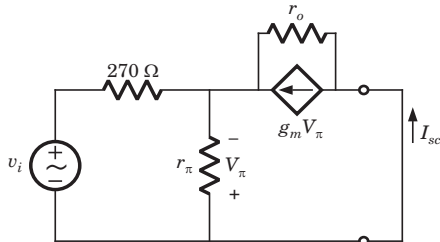


Fig. S 3.4.15

$$I_{sc} = g_m V_\pi + \frac{V_\pi}{r_o} = 2mV_\pi + \frac{V_\pi}{250k} = 2.004 mV_\pi$$

$$\frac{V_\pi}{r_\pi} + \frac{V_\pi}{r_o} + g_m V_\pi + \frac{v_i + V_{pi}}{270} = 0$$

$$\frac{V_\pi}{50k} + \frac{V_\pi}{250k} + 2mV_\pi + \frac{v_i + V_{pi}}{270} = 0 \Rightarrow V_\pi = -0.647 v_i,$$

$$I_{sc} = 1297 mV_i$$

$$R_{TH} = \frac{v_{TH}}{I_{SC}} = \frac{498 v_i}{1297 mV_i} = 384 \text{ k}\Omega$$

16. (B) The equivalent small-signal circuit is shown in fig. S3.4.16

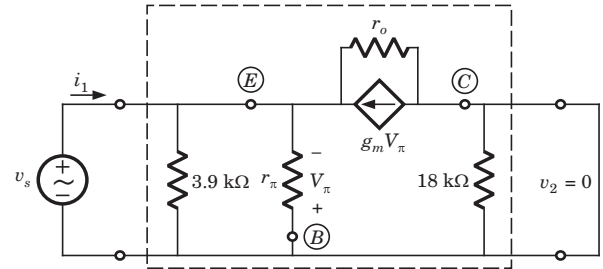


Fig. S 3.4.16

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{3m} = 33.3 \text{ k}\Omega$$

$$h_{21} = \frac{i_2}{i_1} \Big|_{v_2=0}, i_2 = \frac{V_\pi}{r_o} + g_m V_\pi$$

$$i_1 = -\frac{V_\pi}{39k} - \frac{V_\pi}{r_\pi} - \frac{V_\pi}{r_o} - g_m V_\pi, \frac{V_\pi}{r_o} \text{ can be neglected}$$

$$h_{21} = \frac{i_2}{i_1} = \frac{-g_m}{\frac{1}{39k} + \frac{1}{r_\pi} + g_m} = \frac{-g_m r_\pi 39k}{r_\pi + 39k + g_m r_\pi 39k} = 0.91$$

$$17. (A) v_1 = -V_\pi, \frac{v_1}{39k} + \frac{v_1}{r_\pi} + \frac{v_1 - v_2}{r_o} = g_m V_\pi$$

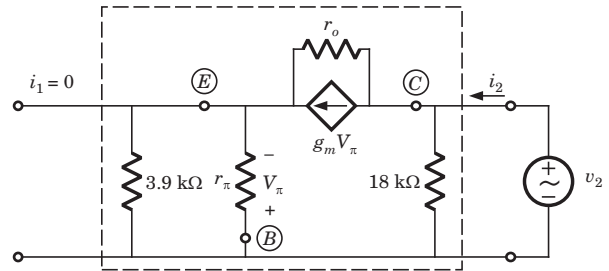


Fig. S 3.4.17

$$v_1 \left(\frac{1}{39k} + \frac{1}{r_\pi} + \frac{1}{r_o} \right) - \frac{v_2}{r_o} = -g_m v_1$$

$$\frac{v_1}{v_2} = \frac{\frac{1}{r_o}}{\frac{1}{39k} + \frac{1}{r_\pi} + \frac{1}{r_o} + g_m} = \frac{\frac{1}{800k}}{\frac{1}{39k} + \frac{1}{33.3k} + \frac{1}{800k} + 3m}$$

$$= 3.8 \times 10^{-4}$$

$$\begin{aligned}
 R_S &= 4 \text{ k}\Omega, v_{gs} = 0.84v_i, \\
 v_o &= -g_m v_{gs} (r_o \parallel R_D) \\
 &= -(1.41\text{m})(0.84v_i)(100\text{k} \parallel 5\text{k}) \\
 \Rightarrow \frac{v_o}{v_i} &= A_v = -5.6
 \end{aligned}$$

28. (A) $R_o = R_D \parallel r_o \parallel 100\text{k} = 4.76 \text{ k}\Omega$

29. (A) As shown in fig. S3.4.27, $R_i = R_1 \parallel R_2 = 20.6 \text{ k}\Omega$

30. (C) From the DC analysis:

$$V_{GSQ} = 1.5 \text{ V}, I_{DQ} = 0.5 \text{ mA}$$

$$g_m = 2K_n(V_{GS} - V_{TN}) = 2(1\text{m})(1.5 - 0.8) = 1.4 \text{ mA/V}$$

$$r_o = [\lambda I_{DQ}]^{-1} = \infty$$

The resulting small-signal equivalent circuit is shown in fig. S5.4.30

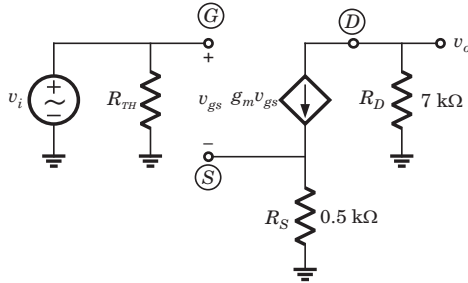


Fig. S3.4.30

$$\begin{aligned}
 v_o &= -g_m v_{gs} R_D, v_i = v_{gs} + g_m v_{gs} R_S \\
 \Rightarrow \frac{v_o}{v_i} &= \frac{-g_m R_D}{1 + g_m R_S} = -(1.4\text{m}) \frac{(7\text{k})}{1 + (1.4\text{m})(0.5\text{k})} = -5.76
 \end{aligned}$$

31. (B) Since the DC gate current is zero, $V_S = -V_{GSQ}$

$$I_{DQ} = I_Q = K_n (V_{GSQ} - V_{TN})^2$$

$$\Rightarrow 0.5 = 1(V_{GSQ} - 0.8)^2$$

$$V_{GSQ} = 1.51 \text{ V} = -V_S$$

$$V_{DSQ} = 5 - (0.5\text{m})(7\text{k}) - (-1.51) = 3.01 \text{ V}$$

The transistor is therefore biased in the saturation region. The small-signal equivalent circuit is shown in fig. S3.4.31.

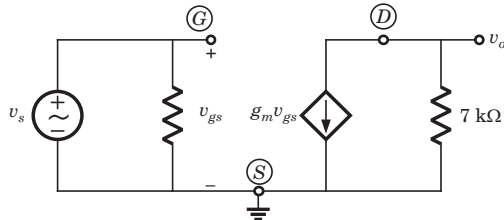


Fig. S3.4.31

$$v_o = -g_m v_{gs} (7\text{k})$$

$$v_{gs} = v_i, \frac{v_o}{v_i} = A_v = -g_m (7\text{k})$$

$$g_m = 2K_n(V_{GS} - V_{TN})$$

$$= 2(1\text{m})(1.51 - 0.8) = 1.42 \text{ mS}$$

$$A_v = -(1.42\text{m})(7\text{k}) = -9.9$$

32. (A) The small-signal equivalent circuit is shown in fig. S3.4.32

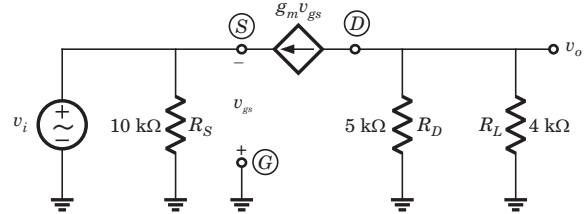


Fig. S3.4.32

$$v_o = -g_m v_{gs} (R_D \parallel R_L), v_i = -v_{gs}$$

$$A_v = \frac{v_o}{v_i} = g_m (R_D \parallel R_L) = (2\text{m})(5\text{k} \parallel 4\text{k}) = 4.44$$

33. (A) The small-signal equivalent circuit is shown in fig. S3.4.33

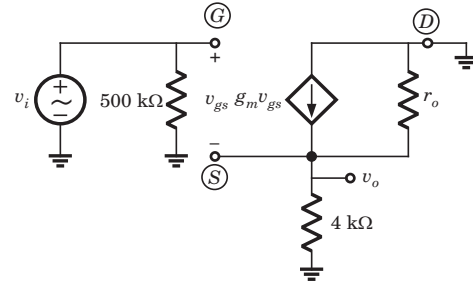


Fig. S3.4.33

$$v_o = g_m v_{gs} (R_L \parallel r_o)$$

$$v_i = v_{gs} + v_o = v_{gs} + g_m v_{gs} (R_L \parallel r_o)$$

$$v_{gs} = \frac{1}{1 + g_m (R_L \parallel r_o)}$$

$$\frac{v_o}{v_i} = A_v = \frac{g_m (R_L \parallel r_o)}{1 + g_m (R_L \parallel r_o)}$$

$$R_L \parallel r_o = 4\text{k} \parallel 100\text{k} = \frac{100\text{k}}{26} = 3.86 \text{ k}\Omega$$

$$A_v = \frac{(2\text{m})(3.85\text{k})}{1 + (2\text{m})(3.85\text{k})} = 0.89$$

34. (B) $R_o = \frac{1}{g_m} \parallel r_o$

$$= \left(\frac{1}{2}\right) \parallel (100) \approx 0.498 \text{ k}\Omega$$
